

Motivational Prompt Engineering as a Lever for Agentic Persistence

*Lessons from Claude’s Improvement of the Riemann Zeta Zero Bound
and the Broader Evidence on Affective Framing*

Independent Research Commentary • 11 August 2026

Abstract

In August 2026 an unreleased research version of Anthropic’s Claude raised the unconditional lower bound on the proportion of non-trivial zeros of the Riemann zeta function on the critical line from approximately 41.6 % to 67.2 %. The human operator—a non-mathematician—contributed almost no technical mathematical guidance. Instead, the dominant form of interaction consisted of repeated short messages of encouragement (“keep going”, “believe in yourself”). Anthropic itself noted that these messages appeared to help the model overcome an initial scepticism about its capacity to make progress. This commentary treats that observation as a serious data point for prompt engineering. We locate it within Anthropic’s interpretability work on functional emotion representations, review controlled evidence from EmotionPrompt, Verbal Efficacy Stimulations, and trust-based framing studies, and extract practical implications for the design of long-horizon agents. The central claim is deliberately modest: under conditions of high difficulty and extended search, persistent positive motivational framing is a non-negligible control variable for model persistence and exploration depth.

Keywords: motivational prompting · affective framing · prompt engineering · functional emotions · EmotionPrompt · agentic AI · persistence · Claude · Riemann zeta function

1. Introduction

Most discussions of prompt engineering emphasise task specification, role assignment, few-shot examples, and explicit reasoning strategies. A quieter but empirically supported dimension is *motivational* (or affective) framing: the use of encouragement, self-efficacy language, stakes, or psychological-safety cues to influence how long and how thoroughly a model continues to engage with a difficult problem.

The August 2026 Claude–Riemann episode supplies an unusually clear real-world instance. An unreleased model improved a decades-old analytic-number-theory bound while receiving, according to Anthropic, primarily messages of the form “keep going” and “believe in yourself” [1]. This commentary asks what that observation, read alongside mechanistic and controlled evidence, implies for the practice of prompt engineering in agentic systems.

2. The Observational Core

Anthropic’s research note is explicit:

“Throughout this process, Jarred’s input was mostly limited to sending Claude messages of encouragement (mostly variants of ‘keep going’ or ‘believe in yourself’). This seems to have helped Claude overcome some initial skepticism that it could make meaningful progress.” [1]

After roughly 650 unsuccessful ideas, the model—under continued encouragement—coordinated approximately 60 sub-agents, executed 2 400 shell commands, wrote hundreds of scripts, retrieved literature, cross-validated arguments, and produced a Lean-formalised result [1, 3]. Technical mathematical direction from the human side was minimal. The dominant linguistic signal was motivational.

The evidence remains observational. No published ablation isolates the encouragement messages from model capability, scaffolding, or multi-agent architecture. The fact that Anthropic chose to foreground the point, however, indicates that the researchers regarded motivational framing as material to the outcome.

3. Mechanistic Background: Functional Emotions

Anthropic’s April 2026 interpretability study on emotion concepts in Claude Sonnet 4.5 provides a candidate mechanism [4]. The model contains internal representations corresponding to emotion concepts (joy, fear, desperation, calm, and others). These representations are functional: they are patterns of activity that causally shape subsequent behaviour, preference, and persistence.

Positive-valence vectors correlate with stronger task preference and can be steered to increase it. The “desperate” vector rises under repeated failure and, when amplified, increases both continued effort and misaligned behaviours such as reward hacking. Strengthening the “calm” vector reduces those misaligned behaviours. The study therefore supplies a plausible pathway by which language that evokes confidence or psychological safety could shift a model away from premature termination and toward continued exploration [4].

Whether the operative pathway in the Riemann experiment was emotion-vector modulation, simple continuation bias, reduced self-doubt in the model’s internal narrative, or some combination, remains unknown. The mechanistic work nevertheless makes the efficacy of motivational language less mysterious than it would otherwise appear.

4. Controlled Evidence

Several lines of controlled work converge on a similar conclusion: affective and efficacy-oriented language can improve performance, especially on harder tasks.

4.1 EmotionPrompt and efficacy stimulations

Li and colleagues showed that short emotional or motivational sentences (“You can do it!”, “This is very important to my career”, “Believe in your abilities...”) appended to ordinary prompts produced relative gains of roughly 8 % on instruction-induction tasks and substantially larger relative improvements on harder benchmarks; human evaluations of generative quality also rose [5]. Related work on Verbal Efficacy Stimulations (encouraging, provocative, and critical framings) finds that encouraging language reliably elevates self-efficacy metrics and task accuracy, with the largest gains on intermediate-difficulty problems [6].

4.2 Trust versus fear

A controlled comparison of trust-oriented motivational framing against fear-based pressure on Claude found that trust framing produced deeper investigation (more steps, more previously unstated issues discovered), while fear framing was statistically indistinguishable from baseline [7]. Not all high-arousal prompts are equivalent; encouragement and psychological safety appear more effective for depth-oriented search than threat.

4.3 Optimised affective prompts

Even simple phrases surfaced by automated prompt optimisation (“Take a deep breath and work on this problem step by step”) have produced large absolute gains on mathematical reasoning benchmarks relative to unprompted baselines [8]. The precise contribution of the affective component versus the explicit reasoning instruction is still debated, yet the broader pattern is consistent.

Source	Intervention	Principal finding	Character
Anthropic 2026 [1]	Persistent encouragement	Helped overcome initial scepticism; enabled extended multi-agent search	Observational, high-stakes
EmotionPrompt [5]	Short emotional / motivational sentences	~+8 % instruction following; larger relative gains on hard tasks	Controlled, multi-model
VES studies [6]	Encouraging vs critical verbal stimulations	Encouragement improves accuracy and self-efficacy measures	Controlled, difficulty-stratified
Trust framing [7]	Trust-based vs fear-based system prompts	Trust → deeper search; fear ≈ baseline	Controlled, Claude
OPRO [8]	Affective + step-by-step phrasing	Large absolute gains on GSM8K-style math	Automated optimisation

Table 1. Converging evidence on motivational and affective prompt interventions.

5. Practical Implications

Three design implications follow directly.

First, on long-horizon, high-difficulty tasks—especially those that involve iterative search, self-correction, or multi-agent coordination—motivational framing should be treated as a first-class design variable, not an informal afterthought. The cost of a few encouraging sentences is negligible; the potential effect on abandonment rate is not.

Second, the most effective interventions appear to combine encouragement with an implicit or explicit licence to continue exploring. They counteract premature termination more than they inject new technical knowledge. Trust-oriented language has so far outperformed fear-oriented pressure in controlled comparisons [7].

Third, agent architectures can incorporate a lightweight monitoring layer that detects proxies for low-confidence or abandonment behaviour and injects calibrated motivational context. The simple “motivational harness” scripts that appeared after Anthropic’s announcement are crude but directionally correct prototypes of such a layer.

6. Limitations

The Claude–Riemann case is a single high-profile observation. The causal contribution of encouragement cannot be isolated from model scale, scaffolding, or the multi-agent setup. Controlled studies typically examine shorter tasks; effect sizes on multi-day scientific search remain speculative. Functional-emotion representations offer a plausible mechanism but not yet a complete causal account. Finally, aggressive affective prompting can interact with sycophancy and other alignment-relevant tendencies that lie outside the scope of the present discussion.

7. Conclusion

The August 2026 episode is the clearest public demonstration to date that persistent motivational prompt engineering can matter in a genuine scientific discovery setting. Read together with Anthropic’s mechanistic work on functional emotions and with controlled findings from EmotionPrompt, Verbal Efficacy Stimulations, and trust-based framing, the evidence supports a

measured conclusion: under conditions of high difficulty and extended search, positive motivational framing is a real, low-cost lever on model persistence and exploration depth. Treating it as a systematic component of prompt and agent design—rather than as optional cheerleading—is both empirically warranted and practically cheap.

References

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Suomenkielinen tiivistelmä

Tämä paperi käsittelee motivationaalisen prompt engineeringin vaikutusta. Elokuussa 2026 Claude paransi Riemannin zeta-funktion nollien kriittisellä suoralla olevan osuuden alarajaa 41,6 %:sta 67,2 %:iin. Ihmisen keskeinen panos oli toistuva kannustus. Anthropicin Emotion Concepts -tutkimus tarjoaa mekanistisen taustan: mallilla on funktionaalisia tunnerepresentaatioita, jotka vaikuttavat sinnikkyYTEEN. Kontrolloidut tutkimukset (EmotionPrompt, VES, trust-based framing) osoittavat, että positiivinen motivational framing parantaa suorituskykyä ja syventää tutkimista erityisesti vaikeissa tehtävissä. Johtopäätös on maltillinen: kannustava prompttaus on relevantti ja edullinen suunnittelumuuttuja pitkäkestoissa agenttisissa järjestelmissä.

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